Avoiding the Edge Before It Is Too Late: When Is Pain a Sign of Injury in Athletes?

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A multitude of internal and external factors converge to produce the output of pain. Although pain is an anticipated side effect of high-intensity exercise, if athletes persist through injury-related pain mistaken for muscle soreness, it may "push them over the edge," leading to injury. The decision to push through or pull back ultimately lies with the athlete. Although athletes are often adept at making this decision, even at the highest level their decision making skills are not infallible, with approximately 9% of athletes at the 2022 Summer Olympics sustaining an injury.[1]

Athletes typically learn the limits of their personal edge and the difference between pain that is a sign of training to one's limit versus injury based on trial-and-error in their decision making.[2] To increase precision in this process, enhancement in the understanding of the meaning of pain (eg, threat value) from the cells to the cortex and how to respond optimally in a given context is required. As a complex array of factors contribute to defining this edge,[3–6] advancement in precisely targeted, multimodal biopsychosocial assessment has the potential to isolate individualized early warning signs.

The BIS-BAS model: A framework for how to be aware of the edge

Evolutionarily, pain demands attention and urges action in response to cues signaling potential harm. Complex processes influence these cues at the level of tissues, afferent input, learning/memory/experience, and socio-environmental context.[3–6] A possible integrative framework for conceptualizing how athletes interpret and respond to pain, supported by electroencephalogram, brain imaging and self-report data, is the Behavioral Inhibition System (BIS)-Behavioral Activation System (BAS) model of behavioral regulation, motivation, and emotion.[5] Although the original BIS–BAS model of pain primarily focuses on internal factors, it has been refined here to incorporate broader spheres of importance to athletes, as depicted in Figure 1.[5] This revised BIS–BAS model acknowledges the significance of environmental information (eg, influence of coaches) and peripheral information (eg, threats to tissues) in pain responses.

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Within this framework, the BIS is activated when we pause in the face of pain, and the BAS is activated when we push through pain to achieve a reward (eg, achieving first place). The activation of the BIS and BAS systems is automatic and effortless, facilitating quick decision-making. Although one's predisposition toward BIS or BAS activation has trait-like characteristics suggesting individual consistency over time, it is modifiable with appropriate interventions.[5] As continued activity despite the presence of stressors increases an athlete's injury risk,[7] an underlying cause of this vulnerability could be a predisposition toward an overactive BAS. In this state, thoughts are focused on reward; emotions are focused on drive; and behaviors are focused on goal-directed action. Although the BAS may appear positive, if these thoughts, emotions, and behaviors are misaligned with contextual cues (eg, from the periphery, such as tissue threat), this may increase injury risk. Moreover, these processes consume cognitive resources, potentially impacting decision making, which again may contribute to alternative input from the periphery not being considered and judgmental error occurring.

With overactivation of the BAS, the default automaticity in response to pain lacks balanced, reflective, precise evaluation whereby the athlete may want to push through the pain, even when that action might cause injury. Accurately judging pain therefore requires a data driven approach to remove any tendency to overload the situation with beliefs, expectancies, emotions, and behaviors that are contextually mismatched. This approach offers the potential to balance BIS and BAS activation, and flexibly shift between these systems in ways that are contextually responsive. To inform this balance, a multi-systems approach to a BIS–BAS complex biopsychosocial profile assessment is needed.

IDENTIFYING WHEN THE EDGE OF INJURY NEARS

To avoid injury when BAS is overactive, when a person is fatigued, or when other vulnerabilities exist, training load and activity need to be pre-emptively modified. We propose evaluative precision is best informed by multimodal biopsychosocial assessment tools to gain real-time information across contexts to inform an individualized athlete profile from the cells to physiological systems/tissues to the cortex (eg, the balance of BIS–BAS and related thinking and behavioral patterns).

At the cellular level, with advances in technology, we are beginning to understand the molecular-level processes that provide information that predict future pain[8] and injury. As cellular processes change tissue biology, blood assays can provide real-time data on when an individual is reaching their edge. Researchers have analyzed the signaling capacity of

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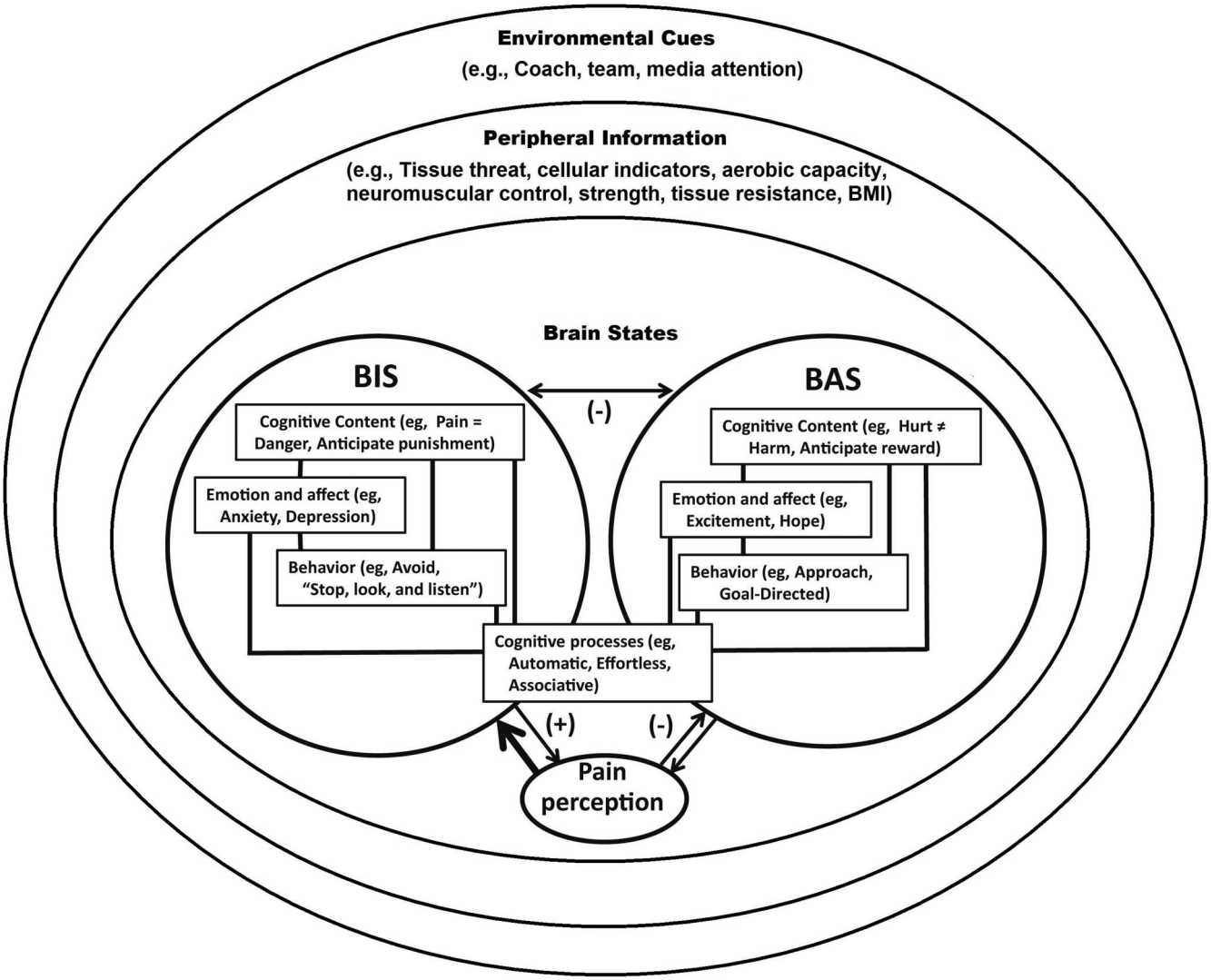

Figure 1. BIS–BAS model of pain adapted for athletes.

Toll-like receptors (TLRs) in the blood. TLRs are an evolutionarily ancient innate immune system pattern recognition receptor system shown to play a key role in the body's defense system by identifying signs of damage or danger.[9,10] Given this critical role, routine blood tests for elite athletes at specified intervals are indicated, and at present, the technology exists to do this weekly.

The experience of pain is also not necessarily linearly related to the biological threat to tissues, thus a physiological metric to judge when the tissues are being pushed toward threat may add further precision. The complex interaction between the applied mechanical loads to the tissue, the individual's physiology, the tissue-specific stress and strain, and environmental factors could influence how the individual responds to training/pain.[3–6] Capturing this complexity requires the use of novel sensors, and research has shown that use of real-time continuous physiological data from wearable sensors (ie, such as heart rate monitors, sleep monitors, and temperature sensors) can improve performance and decrease injury risk.[11] Collection of real-time physiologic data can be done continuously, which is feasible as many athletes already wear such sensors.

Together, these data can be used to further inform cortical processes, including those specified in the BIS–BAS model. From a BIS–BAS perspective, an athlete's thoughts, emotions, and behaviors in response to pain (ie, coping mechanisms) provide essential insights into their psychological state/wellbeing and can be quantified using a variety of tools to characterize typical responses. Although more trait-based characteristics tend to be stable, those state-based characteristics that fluctuate across context need more frequent assessment (such as weekly assessment of affect). To highlight one measure as an example, the Pain Responses Scale[6] assesses each of the key BIS–BAS domains and quantifies an athlete's tendency toward approach- and avoidance-oriented thoughts and behaviors. This information could be used to stratify athletes, for example, by whether they have an overactive or underactive BIS or BAS.

An athlete's response to pain is also not functioning in isolation from socioenvironmental cues. All human behavior, including perception of injury risk, operates interactionally with the wider social context. Factors such as criticism from coaches and teammates,[12] the team's culture of acceptance of playing through pain/injury, and performance pressure,[13] all

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influence an athlete's response to pain. Such factors can drive athletes to ignore their pain signals, pushing their body beyond safe limits.[14] Therefore, a comprehensive biopsychosocial assessment should also include a robust consideration of these social elements, potentially through conducting interviews with athletes and staff. Given social and environmental factors tend to remain stable, frequent assessment would not be indicated. However, understanding these dynamics within the athlete's context, and their role in shaping an athlete's perception of injury and response to pain, offers insight and provides a potential pathway to address such elements within an athlete's environment.

An example biopsychosocial assessment battery that can be tailored to individual athletes is provided in **Supplemental Digital Content 1** (see **Table**, http://links.lww.com/JSM/A418). By collating and integrating this cells-to-cortex data profile, we can potentially assess whether an athlete is nearing their personal proximity to the edge of injury. Although we acknowledge that implementation of these measures is intensive, we argue that they are feasible for elite athletes. The requisite blood tests could be obtained at a relatively minimal cost, as low as $67 AUD per test. Furthermore, the required physiological measurements are typically readily accessible to athletes competing at a high level. Although there may be concerns about athletes becoming reliant on these tests, we argue that by enhancing athletes' ability to make informed judgements about their pain, they will be able to better interpret their pain signals, thereby contributing to *less* reliance on or need for testing over time. This small investment of time and resources could be instrumental in preventing injuries, preserving athletes' long-term health/well-being, and maintaining optimal performance.

CONCLUSION

We propose that an assessment of peripheral information at the cellular to tissue level, integrated with a BIS–BAS assessment of cortical level processes is critical to define "the edge." This approach could precisely inform a data-driven decision to modify the athlete's BIS–BAS response. By using this multimodal data that is repeatedly obtained in a time-series fashion to inform cortex-level processes, the accuracy of decision-making will no longer rely on trial-and-error, but precise algorithms uniquely formulated for a given individual in their current context. Athletes will then have the potential to strike a balance between BIS and BAS activation and flexibly shift between these systems in a way that is contextually responsive. This holistic perspective can help to optimize how athletes judge and respond to pain in a way that avoids both undertraining and overtraining to perform at their peak, when it matters most.

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